Junxiang Huang

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Education

Peking University Sep. 2022 – Expected Jun. 2027
PhD Candidate in Computer Software and Theory *Beijing, China*

Peking University Sep. 2018 – Jul. 2022
B.S. in Physics and B.S. in Computer Science *Beijing, China*

Research Experience

Xiao Yuan's Group Apr. 2021 – Present
Center on Frontiers of Computing Studies, Peking University *Beijing, China*
Research in Quantum Information and Quantum Computing

- Developed resource-efficient quantum algorithms and theoretical tools for many-body simulation and information extraction, spanning variational, tensor-network-assisted, and early fault-tolerant approaches.
- Designed qubit-, measurement-, and sample-efficient protocols for estimating local observables and nonlinear quantum-state properties, including partial-transpose moments.
- Developed a correlation-threshold criterion for selecting the scope of error mitigation in collaborative large-scale cluster-state experiments (Nature Physics, 2026).
- Current projects include response-transform and counterdiabatic algorithms, fault-tolerant resource estimation, and neural-network quantum states for open-system dynamics.

Yuanjie Exploration Sep. 2026 – Present
Research Intern in Quantum Error Correction

- Participating in quantum error correction research and product development.

Quantum Information Theory Group at Imperial College London Jul. – Aug. 2023
Electrical and Electronic Engineering Department, Imperial College London *London, UK*
Visiting Research Student

- Studied applications of perturbative quantum simulation to quantum information and many-body dynamics.

Research Interests

- Resource-efficient quantum algorithms for many-body simulation, eigenstate and dynamical problems, with particular emphasis on the early fault-tolerant regime.
- QEC/QEM-aware estimation protocols, algorithm–architecture resource analysis, and hybrid quantum–classical methods for quantum simulation.

Publications & Preprints

[10] **J. Huang**, C. Li, L.-F. Zhang, et al. Breaking the Multiplicative Overhead in Quantum Entropy Estimation. arXiv:2609.40179, 2026.

[9] Y. Tang*, Z. Huo*, **J. Huang***, et al. Low-cost algorithm-to-execution framework for surface-code quantum computing. arXiv:2609.10965, 2026.
*Equal contribution.

[8] J. Cai*, **J. Huang***, F. Otto*, Y. Li*, et al. Generalized Mermin Inequalities for Benchmarking Large-Scale GHZ States. arXiv:2607.23574, 2026.
*Equal contribution.

[7] Z. Wang*, **J. Huang***, R. Ye*, et al. A Review of Variational Quantum Algorithms: Insights into Fault-Tolerant Quantum Computing. Front. Phys. 22, 023301 (2027); published online Jul. 2026.
*Equal contribution.

[6] J. Sun*, B. Zhou*, J. Xu*, Y. Yao*, Z. Du*, Z. Zhang*, Y. Gu*, **J. Huang**, et al. Quantum-classical crossover in fault-tolerant quantum dynamics simulation. arXiv:2607.16116, 2026.
*Equal contribution.

[5] **J. Huang**, X. Wang, X. Yuan, Y. Zhang. Streaming the partial-transpose moment hierarchy with order-independent quantum memory. arXiv:2606.14204, 2026.

[4] Q.-M. Ding, J. Peng, **J. Huang**, et al. Obtaining Accurate Ground-State Properties on Near-term Quantum Devices. Phys. Rev. Applied 25, 054057 (2026).

[3] T. Jiang*, J. Cai*, **J. Huang***, N. Zhou*, Y. Zhang, et al. One- and two-dimensional cluster states for topological phase simulation and measurement-based quantum computation. Nat. Phys. 22, 430–438 (2026).
*Equal contribution.

[2] **J. Huang**, Y. Tang, X. Yuan. Estimating Local Observables via Cluster-Level Light-Cone Decomposition. Phys. Rev. A 113, 032414 (2026).

[1] **J. Huang***, W. He*, Y. Zhang, et al. Tensor-Network-Assisted Variational Quantum Algorithm. Phys. Rev. A 108, 052407 (2023).
*Equal contribution.

Invited Talks & Seminars

Resource-Efficient Quantum Property Estimation: Partial-Transpose Moments and Quantum Fisher Information
The University of Tokyo *Oct. 2026*

Estimating Local Observables via Cluster-Level Light-Cone Decomposition
Nanjing University (online) *Dec. 2025*

Efficient Estimation of Global and Local Observables in Clustered Quantum Systems
Shenzhen International Quantum Academy *Jul. 2025*

Referee Service

Journal referee service: Physical Review Letters, Physical Review A, Physical Review B, npj Quantum Information, Quantum
Conference referee service: AQIS, TQC

Awards & Honors

National Scholarship for Doctoral Students
Peking University, 2025–2026 Academic Year — *2026*

Merit Student
Peking University, 2025–2026 Academic Year — *2026*

Award for Scientific Research, 2024–2025 Academic Year
Peking University — *Dec. 2025*

Special Prize (Team Award; Project Lead), 19th "Challenge Cup"
2025 China Youth Science and Technology Innovation Challenge Track — *Nov. 2025*

3rd Place Award, Student Poster Presentation
AQIS 2023 Seoul — *Sep. 2023*

Award for Scientific Research, 2019–2020 Academic Year
Peking University — *Dec. 2020*

Award for Academic Excellence, 2018–2019 Academic Year
Peking University — *Dec. 2019*

First Prize, National Final Round; Selected for the National Training Team
34th Chinese Physics Olympiad — *Nov. 2017*

Selected Funded Projects

Participant, Beijing Natural Science Foundation: Youth Project on dynamical quantum phase transitions (2026–present); Interdisciplinary Key Project on practical quantum information processing (2025–present).

Patents

- X. Yuan, **J. Huang**. Method and Apparatus for Quantum Variational Molecular Dynamics Simulation. CN119811513B, granted Jul. 2025.
- X. Yuan, **J. Huang**. Hybrid Computing Method and Apparatus Based on Neural Network and Quantum Circuit. CN114742228B, granted May 2023.

Poster Presentations

Estimating Local Observables via Cluster-Level Light-Cone Decomposition
AQIS 2026 Daejeon — *Aug. 2026*

Simultaneous Estimation of Partial-Transpose Moments with Active Memory Independent of the Moment Order
AQIS 2026 Daejeon — *Aug. 2026*

Efficient Estimation of Global and Local Observables in Clustered Quantum Systems
AQIS 2025 Hong Kong — *Aug. 2025*

Error Mitigation via N-Representability conditions
TQC 2024 Okinawa and SEEQA 2024 Oxford — *Sep. 2024*

Tensor-network assisted variational quantum algorithm
AQIS 2023 Seoul — *Aug. 2023*

Teaching and Mentoring Experience

Peking University (Class Advisor / Undergraduate Cohort Counselor)
EECS Class 10 (Cohort of 2022) — *Sep. 2022 – Jul. 2026*

Peking University (Teaching Assistant)
Quantum Information — *2023 Spring*

Peking University (Teaching Assistant)
Probability and Statistics (A) — *2022 Spring*

Specialized Skills

Languages: Chinese Mandarin (native speaker), English (fluent)
Programming Languages: Python (mostly used), C++ (familiar), Julia (beginner)